

Dataset Proposal — Phase 1 Submission

Course Project: End-to-End Machine Learning

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Dataset Overview

The chosen dataset is the **UCI Bike Sharing Dataset**, publicly available at the UCI Machine Learning Repository (<https://archive.ics.uci.edu/dataset/275/bike+sharing+dataset>). It contains hourly bike rental records from the Capital Bikeshare system in Washington D.C., covering the years 2011 and 2012. The hourly version of the dataset — which will be used in this project — contains 17,379 rows and 17 columns, making it substantially above the 500-row minimum requirement. All data is fully public and contains no private or personal information. It will be used as a static CSV file downloaded directly from the UCI repository.

Features

The dataset includes a rich mix of numeric and categorical features:

- **Categorical/ordinal:** season, year, month, hour, holiday, weekday, working day, weather situation
- **Numeric:** normalized temperature, normalized "feels-like" temperature, normalized humidity, normalized windspeed
- **Target columns available:** registered users count, casual users count, and total rental count (cnt)

This mix of feature types satisfies the course requirement for datasets with both numeric and categorical variables.

Classification Question

"Will bike demand in a given hour be high?"

A binary categorical target variable `demand_level` will be derived from the existing `cnt` column. Observations above the 70th percentile of hourly rental counts will be labeled `high`; all others will be labeled `low`. This derivation follows the same logic as the worked example in the course (where the customer return label was derived from transaction history) and allows a meaningful classification task without introducing data leakage.

Regression Question

"How many bikes will be rented in a given hour?"

The numeric target is the existing `cnt` column, which records the total number of bike rentals per hour. This is a direct, naturally available regression target that does not require any derivation. The goal is to predict exact rental demand given weather and calendar conditions.

Why This Dataset Is Interesting

Urban bike-sharing systems must balance bike availability, station capacity, and operational staffing on an hourly basis. Accurately predicting both whether a given hour will be high-demand (classification) and how many bikes will actually be needed (regression) has direct real-world implications for city transport planning and sustainability goals. The dataset captures the interaction between weather, time of day, and human behaviour in a transportation context — making the results interpretable and practically meaningful.

Static vs. Live Data — and a Potential Upgrade

For this project, the dataset will be used as a **static CSV file** to ensure full reproducibility and leakage-safe preprocessing, as required by the course. This is the simplest and most reliable approach for a controlled ML study.

However, one clear upgrade path exists: since weather conditions (temperature, humidity, windspeed, weather situation) are among the most predictive features in this dataset, a natural enhancement would be to **connect the deployed Streamlit app to a live weather API** — such as the Open-Meteo or OpenWeatherMap API — so that the app automatically retrieves current weather for a chosen city and feeds it into the trained model to generate real-time predictions. This would transform the static prediction demo into a genuinely live, location-aware tool, without changing the underlying trained model or pipeline.